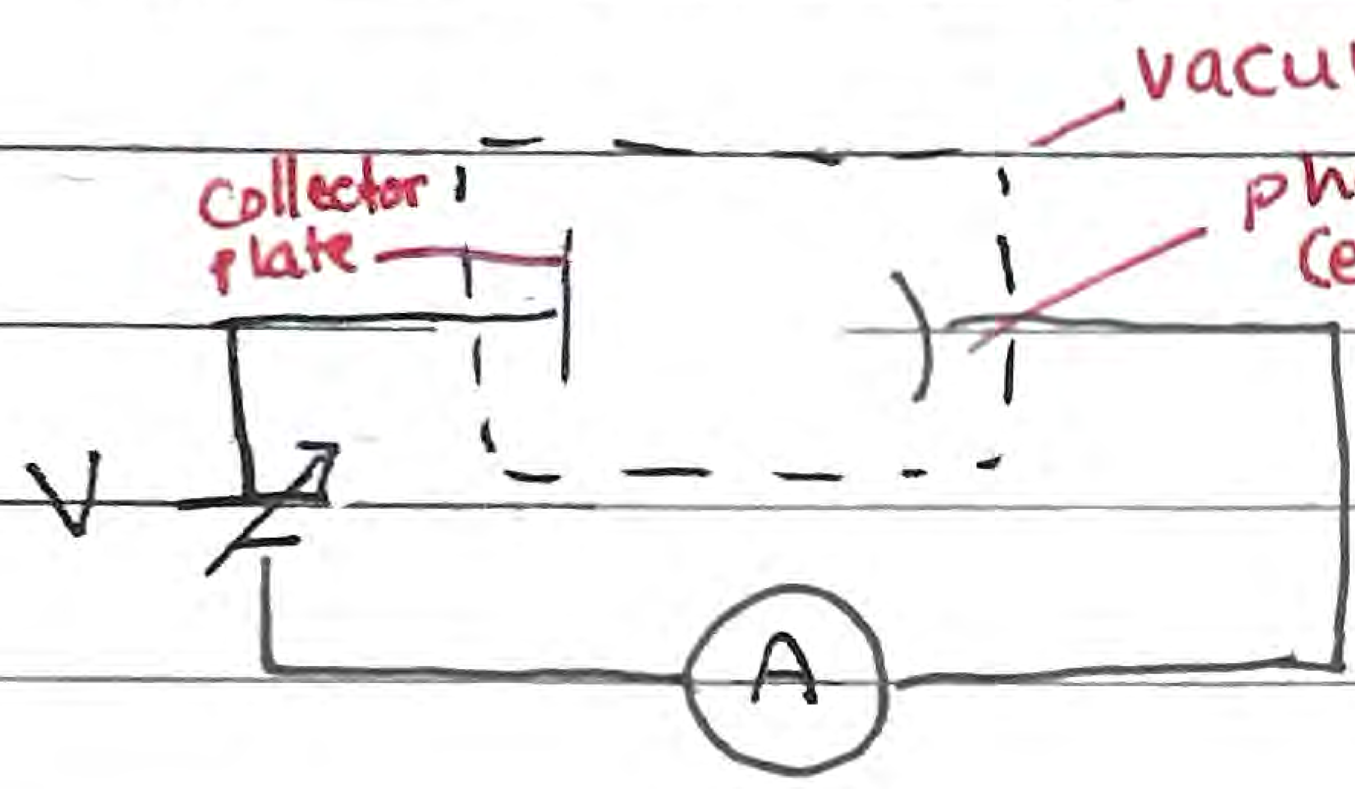
Date. NO.

Photoelectric Effect

Light, when shone at an open circuit with a certain wavelength, can "complete" a circuit, allowing current to pass through

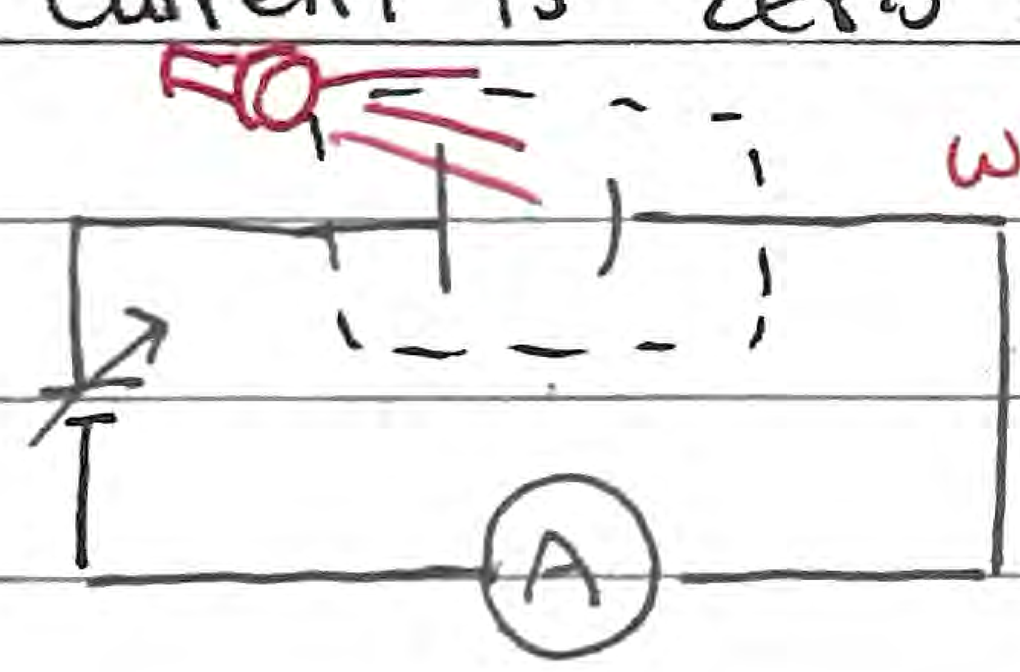
↳ Discovered in 1800s

↳ The circuit is as follows



$I = 0$

The current is zero because circuit is open

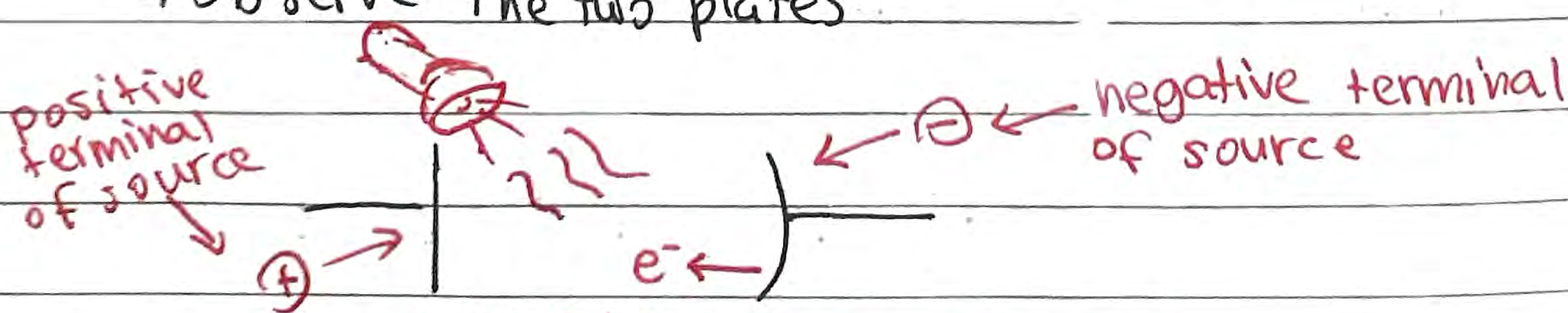


when light is shone, current starts passing through

$I > 0$

But why? Old theory : Wave theory predictions

↳ Observe the two plates



as light hits photo surface, it ejects electrons

· In a vacuum, electrons face no obstacle

↳ Paired with charged plates, the field forces electrons to complete circuit

Photoelectric Effect

This explanation is based on classical wave theories ⇒ Step-by-step process indicates time delay

↳ The scenario happens as long as energy is sufficient to eject electrons

↳ Hence, any wavelength works, as long as amplitude and intensity is enough ($E \propto A$)

↳ E_k of electrons gets bigger as the amplitude (brightness) increases

But the issue is, this phenomenon only happens for select wavelengths!

Photon Model Predictions (Einstein)

↳ 1905, based on light as photon

↳ Photon - Smallest bundle or quonta of Electromagnetic radiation

↳ A cutoff frequency of light is required in order to eject the electrons from the emitter plate

↳ New theory means frequency of light matters

↳ Brighter light means more electrons are ejected but does not affect E_k

↳ E_k is independent of amplitude. Instead, $A \propto$ # of e^-

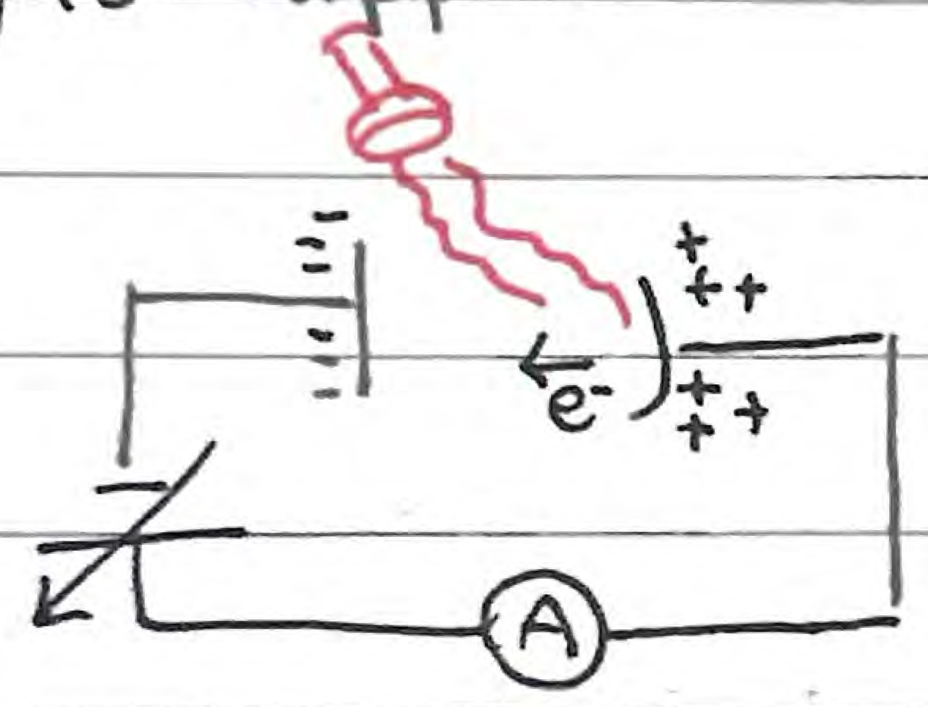
↳ Higher frequency of light produces more energetic electrons

↳ $f \propto E_k$

Photon Model Predictions

- Additionally, there is no time delay in the phenomenon
  - ↳ After light hits photosurface, current begins instantly
  - ↳ Energy transferred instantly, contrasting old wave theories

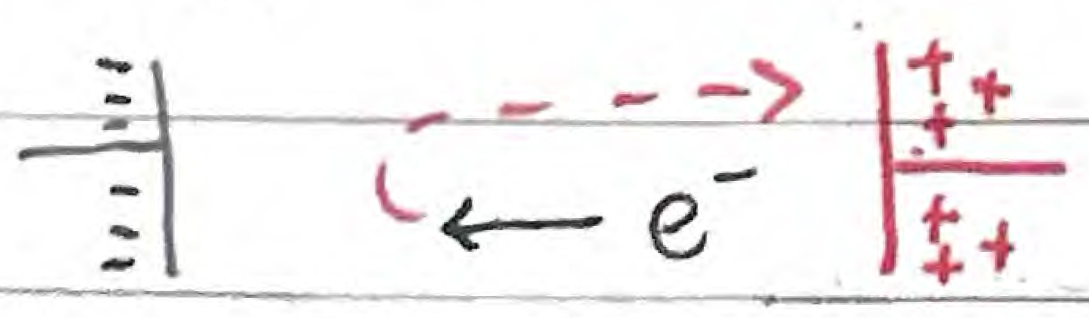
Let's consider the case where the voltage source polarity is flipped



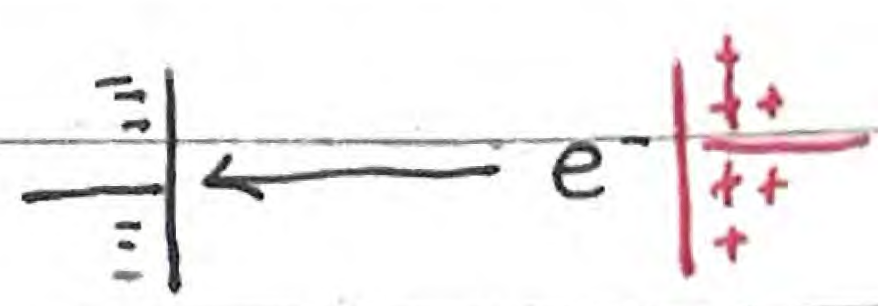
e^- are ejected in all directions (so probabilistically, the wanted direction must also exist)

The photons transfer energy to the photosurface material
↳ It first overcomes energy to eject electrons (> Ionization E)
↳ The excess energy is reflected as kinetic energy

But, since polarity is flipped, e^-'s are repulsed from collector plate, so kinetic energy must also be enough for current to flow



insufficient



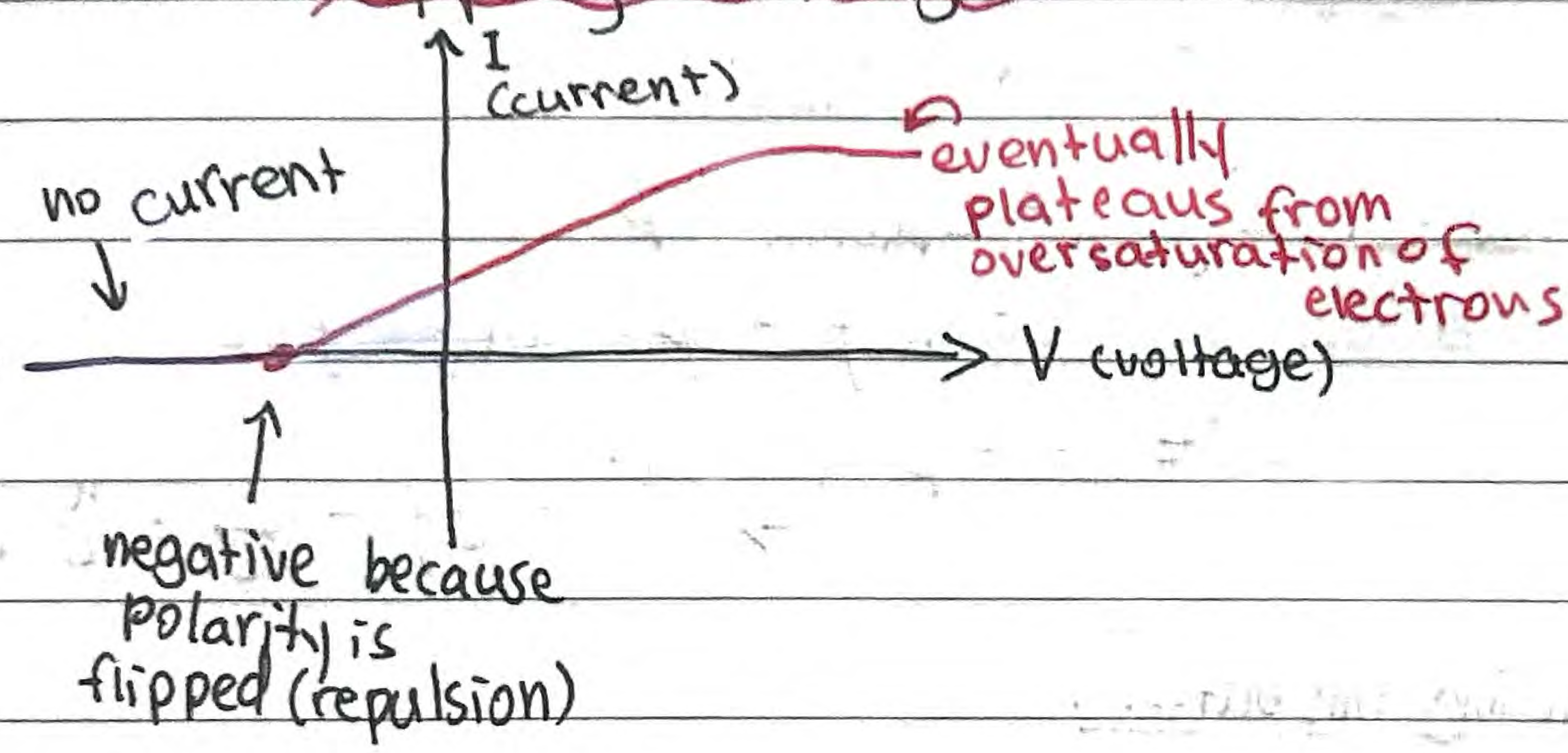
enough

Photoelectric effect

But we have a variable voltage source

↳ That potential difference controls the repulsion the electron experiences

↳ The minimum voltage at which there exists current (electrons barely complete circuit) is the Stopping Voltage



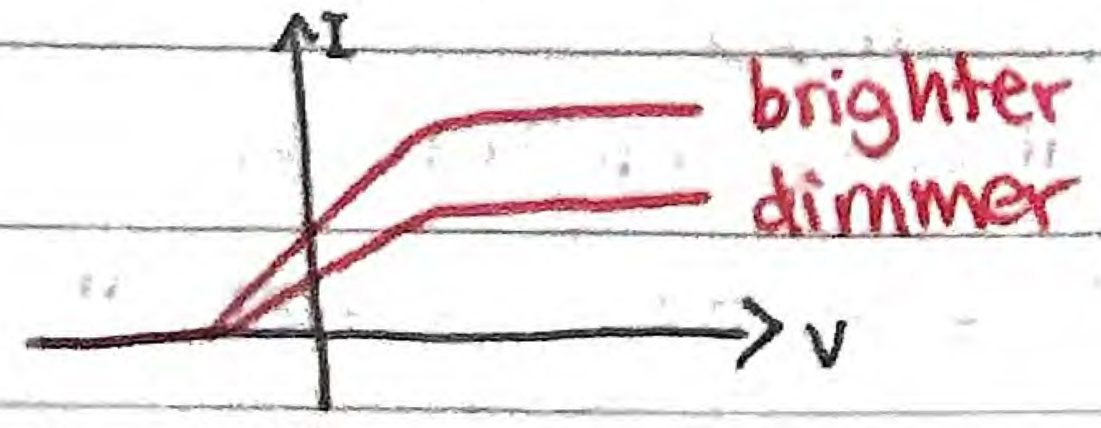
Stopping voltage is the minimum potential difference

↳ Hence maximum kinetic energy is given by

$$\boxed{E_k = qV \Rightarrow E_{k,max} = e \cdot V_s}$$

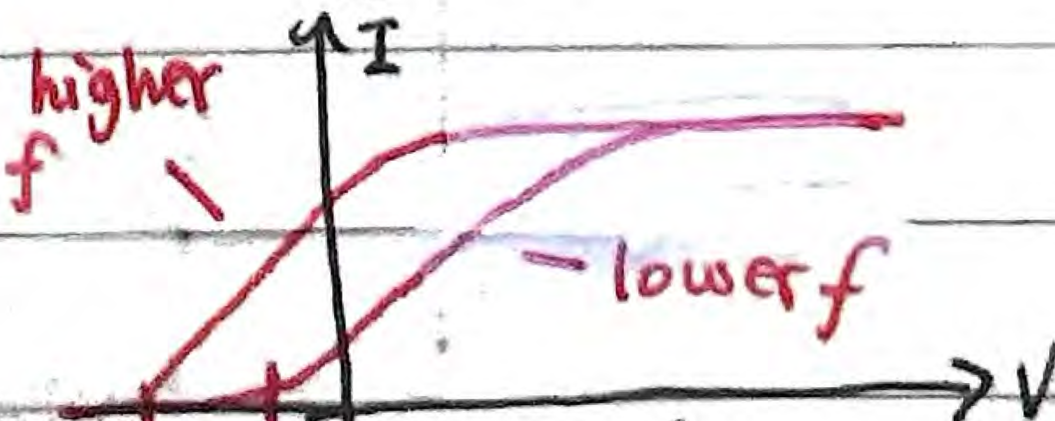
We've changed properties of the voltage source, but how about properties of light and photons?

Current vs Voltage
(varying intensity)



- Same V_s
- Greater slope/max. for brighter (more e^- ejected)

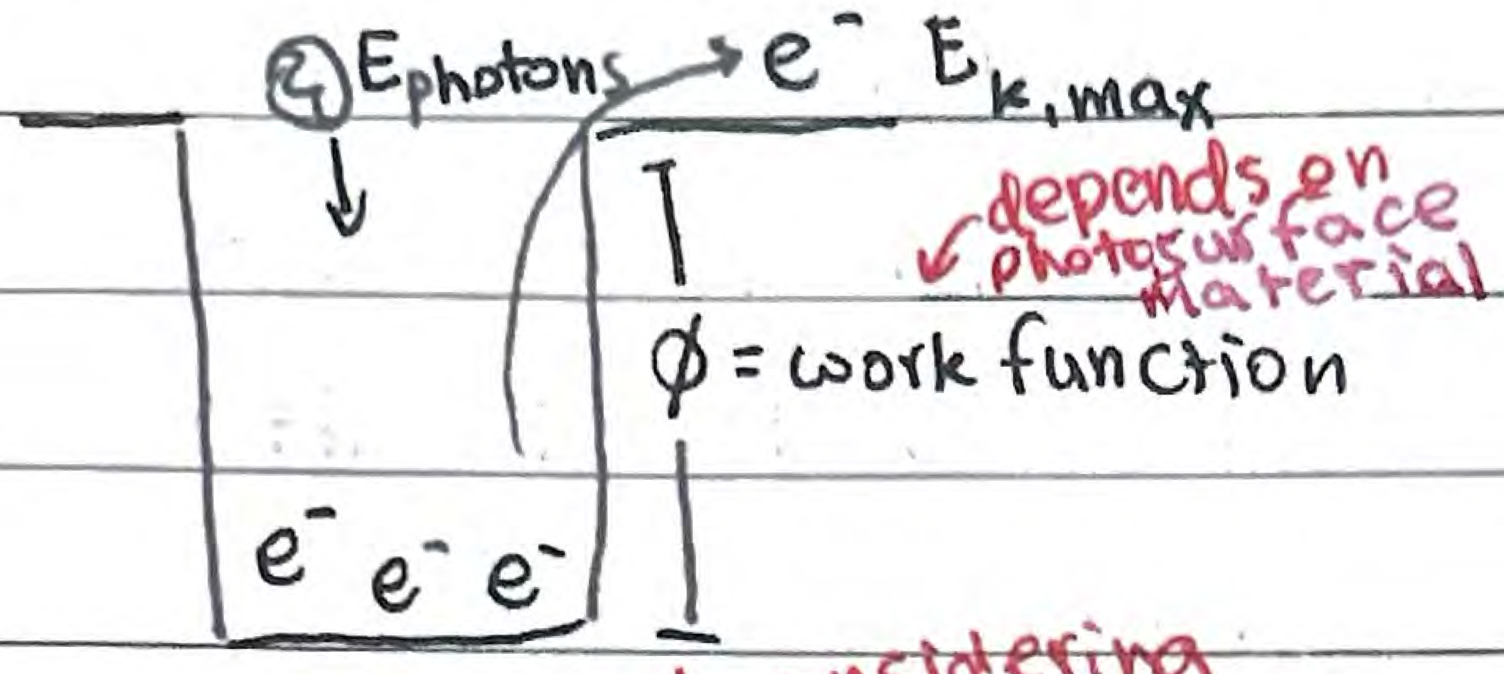
Current vs Voltage
(varying frequency)



- Different V_s (E_k different)
- But same max/slope (same # of e^- ejected)

Photoelectric Effect

observing the photosurface more specifically...



work function needs to be overcome before e^- are free

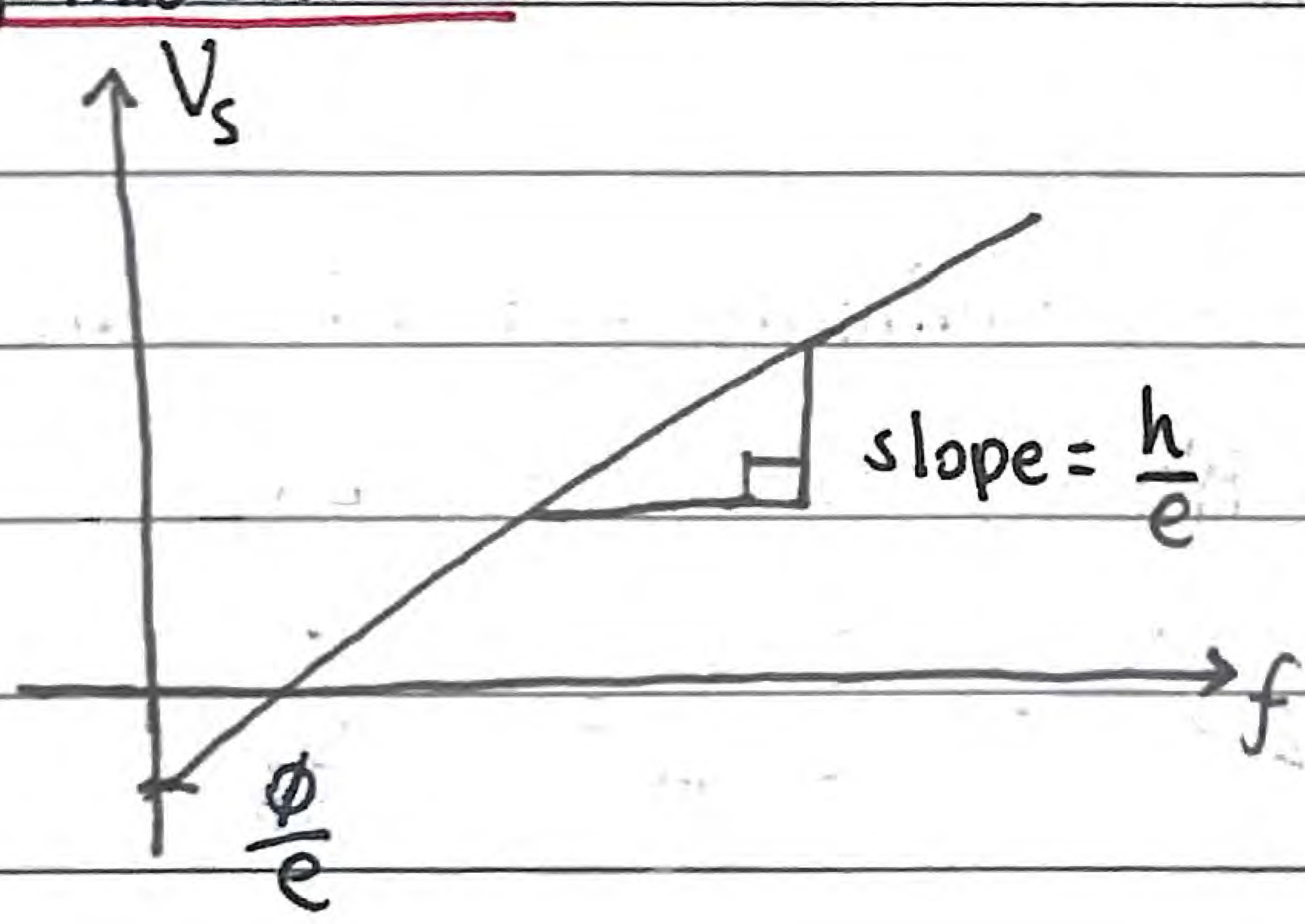
only considering photosurface, no energy gained after

Hence... $E_{k\,max} = E_{photon} - \phi$

$$= h \cdot f - \phi = h(f - f_0)$$

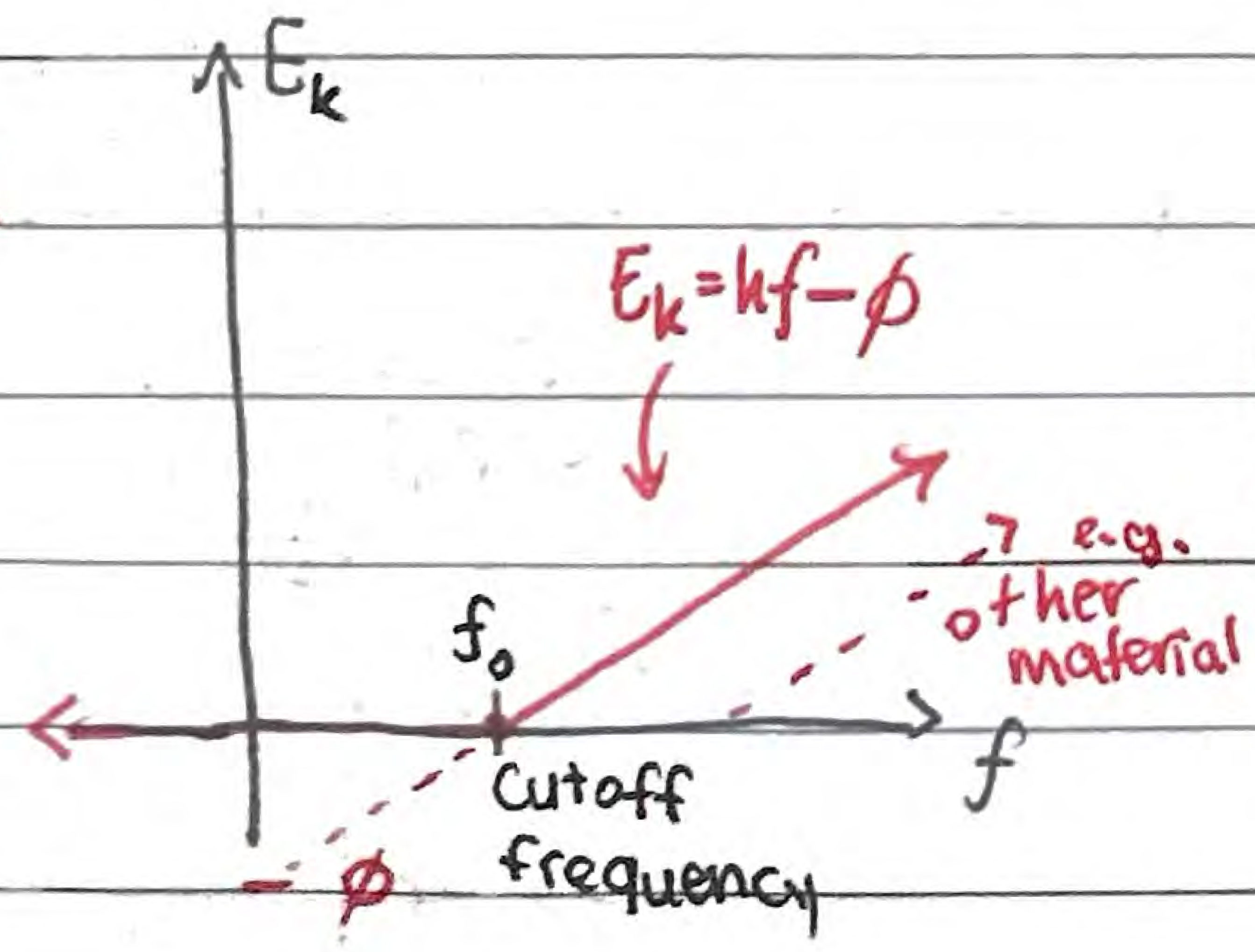
$$eV_s = hf - \phi \rightarrow V_s = \frac{h}{e} f - \frac{\phi}{e} = \frac{h}{e}(f - f_0)$$

Graphing this out...



Sometimes power can be given
↳ You get energy per second
↳ And photons per second

and...



Different materials have different cutoff frequencies (different ϕ)
↳ But every material has same slope (planck's constant h)

this is where the constant is obtained

Photoelectric Effect

E.g. A copper plate is illuminated by a beam of monochromatic light of wavelength 580 nm and power 0.5mW. The work function of Copper is 4.7eV

a) Calculate the # of photons incident on the copper plate in one second

$E_{photon} \approx 3.4 \times 10^{-19}$ J $\Rightarrow P_{total} = 0.5 \times 10^{-3}$ J s^{-1}

So # photons /s $= P_{total} \div E_{photon} \approx 1.5 \times 10^{15}$ photons s^{-1}

b) Show that no electrons will be emitted

$E_{photon} \approx 2.1$ eV < work function ϕ